

Measures of Central Tendency

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Summary Statistics

- In the previous chapter, we constructed tables from raw data.
- The resulting “pictures” of frequency distributions illustrated trends and patterns in the data.
- In most cases, however, we need more exact measures. In these cases, we can use single numbers called summary statistics to describe the characteristics of a data set.
- Two of these characteristics are particularly important to decision-makers:
 - Central tendency
 - Dispersion
- Central Tendency: Central tendency is the middle point of a distribution. Measures of central tendency are also called measures of location.
- Dispersion: Dispersion is the spread of the data in a distribution, that is, the extent to which the observations are scattered.
- There are two other characteristics of data sets that provide useful information: skewness and kurtosis.

Measure of Central Tendency: Arithmetic Mean

- Also called average.
- To find the arithmetic mean, we sum the values and divide by the number of observations.
- Table 3-1 presents data describing the number of days the generators at a power station are out of service owing to regular maintenance or some malfunction.

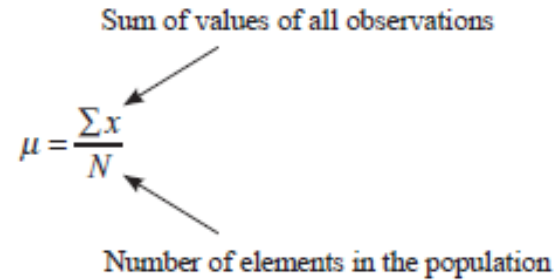
TABLE 3-1 DOWNTIME OF GENERATORS AT LAKE ICO STATION

GENERATOR	1	2	3	4	5	6	7	8	9	10
DAYS OUT OF SERVICE	7	23	4	8	2	12	6	13	9	4

$$\begin{aligned}\text{Arithmetic mean} &= \frac{7 + 23 + 4 + 8 + 2 + 12 + 6 + 13 + 9 + 4}{10} \\ &= \frac{88}{10} \\ &= 8.8 \text{ days}\end{aligned}$$

Calculating the Mean from Ungrouped Data

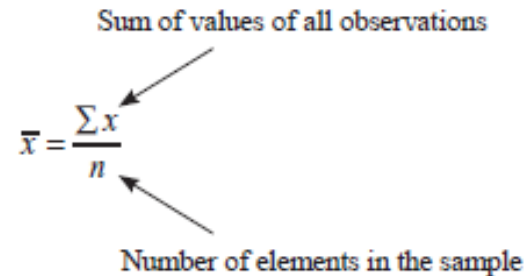
- Population arithmetic mean

$$\mu = \frac{\sum x}{N}$$


Sum of values of all observations

Number of elements in the population

- Sample arithmetic mean

$$\bar{x} = \frac{\sum x}{n}$$


Sum of values of all observations

Number of elements in the sample

- Table 3-2 lists the percentile increase in SAT verbal scores shown by seven different students taking an SAT preparatory course.

TABLE 3-2 SAT VERBAL SCORES

STUDENT	1	2	3	4	5	6	7
INCREASE	9	7	7	6	4	4	2

Calculating the Mean from Grouped Data

- To find the arithmetic mean of grouped data we first calculate the midpoint of each class. Then we multiply each midpoint by the frequency of observations in that class, sum all these results and divide the sum by the total number of observations in the sample. The formula looks like this:

$$\bar{x} = \frac{\sum(f \times x)}{n}$$

TABLE 3-3 AVERAGE MONTHLY BALANCE OF 600 CUSTOMERS

Class (Dollars)	Frequency
0–49.99	78
50.00–99.99	123
100.00–149.99	187
150.00–199.99	82
200.00–249.99	51
250.00–299.99	47
300.00–349.99	13
350.00–399.99	9
400.00–449.99	6
450.00–499.99	4
	600

TABLE 3-4 CALCULATION OF ARITHMETIC SAMPLE MEAN FROM GROUPED DATA IN TABLE 3-3

Class (Dollars) (1)	Midpoint (x) (2)		Frequency (f) (3)		$f \times x$ (3) $\times$ (2)
0-49.99	25.00	$\times$	78	=	1,950
50.00-99.99	75.00	$\times$	123	=	9,225
100.00-149.99	125.00	$\times$	187	=	23,375
150.00-199.99	175.00	$\times$	82	=	14,350
200.00-249.99	225.00	$\times$	51	=	11,475
250.00-299.99	275.00	$\times$	47	=	12,925
300.00-349.00	325.00	$\times$	13	=	4,225
350.00-399.99	375.00	$\times$	9	=	3,375
400.00-449.99	425.00	$\times$	6	=	2,550
450.00-499.99	475.00	$\times$	4	=	1,900
			$\Sigma f = n = 600$		$85,350 \leftarrow \Sigma(f \times x)$
			$\bar{x} = \frac{\Sigma(f \times x)}{n}$		[3-3]
			$= \frac{85,350}{600}$		
			$= 142.25 \leftarrow$	Sample mean (dollars)	

- Using a technique called coding, we eliminate the problem of large or inconvenient midpoints.
- Instead of using the actual midpoints to perform our calculations, we can assign small-value consecutive integers (whole numbers) called codes to each of the midpoints.
- We will assign zero to the midpoint in the middle (or the one nearest to the middle) of the frequency distribution. Then we can assign negative integers to values smaller than that midpoint and positive integers to those larger.
- Symbolically, statisticians use x_0 to represent the midpoint that is assigned the code 0, and u for the coded midpoint.
- The following formula is used to determine the sample mean using codes:

$$\bar{x} = x_0 + w \frac{\sum(u \times f)}{n}$$

w = numerical width of the class interval

TABLE 3-5 ANNUAL SNOWFALL IN HARLAN, KENTUCKY

Class (1)	Midpoint (x) (2)	Code (u) (3)		Frequency (f) (4)		$u \times f$ (3) $\times$ (4)
0-7	3.5	-2	$\times$	2	=	-4
8-15	11.5	-1	$\times$	6	=	-6
16-23	$19.5 \leftarrow x_0$	0	$\times$	3	=	0
24-31	27.5	1	$\times$	5	=	5
32-39	35.5	2	$\times$	2	=	4
40-47	43.5	3	$\times$	2	=	6
				$\Sigma f = n = 20$		$5 \leftarrow \Sigma(u \times f)$

$$\bar{x} = x_0 + w \frac{\Sigma(u \times f)}{n} \quad [3-4]$$

$$= 19.5 + 8 \left(\frac{5}{20} \right)$$

$$= 19.5 + 2 = 21.5 \leftarrow \text{Average annual snowfall}$$

- The frequency distribution below represents the weights in pounds of a sample of packages carried last month by a small airfreight company.

Class	Frequency
10.0–10.9	1
11.0–11.9	4
12.0–12.9	6
13.0–13.9	8
14.0–14.9	12

Class	Frequency
15.0–15.9	11
16.0–16.9	8
17.0–17.9	7
18.0–18.9	6
19.0–19.9	2

- Compute the sample mean.
- Compute the sample mean using the coding method with 0 assigned to the fourth class.

Advantages and Disadvantage of the Arithmetic Mean

- Advantages:
 - Its concept is familiar to most people and intuitively clear.
 - Every data set has a mean. It is a measure that can be calculated, and it is unique because every data set has one and only one mean.
 - The mean is useful for performing statistical procedures such as comparing the means from several data sets.
- Disadvantage:
 - Although the mean is reliable in that it reflects all the values in the dataset, it may also be affected by extreme values that are not representative of the rest of the data.
 - It is tedious to compute the mean because we do use every data point in our calculation (unless, of course, we take the short-cut method of using grouped data to approximate the mean).
 - We are unable to compute the mean for a data set that has open-ended classes at either the high or low end of the scale.

Weighted Mean

- The weighted mean enables us to calculate an average that takes into account the importance of each value to the overall total.

TABLE 3-8 LABOR INPUT IN MANUFACTURING PROCESS

Grade of Labor	Hourly Wage (x)	Labor Hours per Unit of Output	
		Product 1	Product 2
Unskilled	\$5.00	1	4
Semiskilled	7.00	2	3
Skilled	9.00	5	3

- The average cost of labor per hour- \$7
- The labor cost of one unit of product1 to be $\$7(1 + 2 + 5) = \56 and of one unit of product2 to be $\$7(4 + 3 + 3) = \70 , but these answers are incorrect.
- For product1, the total labor cost per unit is $(\$5 \times 1) + (\$7 \times 2) + (\$9 \times 5) = \64 , and, since there are 8 hours of labor input, the average labor cost per hour is $\$64/8 = \8.00 per hour.

- We weight the hourly wage for each grade by its proportion of the total labor required to produce the product.
- One unit of product 1, for example, requires 8 hours of labor.
- Unskilled labor uses 1/8 of this time, semiskilled labor uses 2/8 of this time, and skilled labor requires 5/8 of this time.
- If we use these fractions as our weights, then one hour of labor for product 1 costs an average of

$$\left(\frac{1}{8} \times \$5\right) + \left(\frac{2}{8} \times \$7\right) + \left(\frac{5}{8} \times \$9\right) = \$8.00 / \text{hour}$$

- Hence, the formula for weighted average is:

$$\bar{x}_w = \frac{\sum(w \times x)}{\sum w}$$

- Dave's Giveaway Store advertises, "If our average prices are not equal or lower than everyone else's, you get it free." One of Dave's customers came into the store one day and threw on the counter bills of sale for six items she bought from a competitor for an average price less than Dave's. The items cost

\$1.29 \$2.97 \$3.49 \$5.00 \$7.50 \$10.95

- Dave's prices for the same six items are \$1.35, \$2.89, \$3.19, \$4.98, \$7.59 and \$11.50. Dave told the customer, "My ad refers to a weighted average price of these items. Our average is lower because our sales of these items have been:"

7 9 12 8 6 3

- Is Dave getting himself into or out of trouble by talking about weighted averages?

Arithmetic Mean

Simple average

Weighted Mean

Values with different
importance

Geometric Mean

Growth rates, ratios

Geometric Mean

- when we are dealing with quantities that change over a period of time, we need to know an average rate of change, such as an average growth rate over a period of several years.
- In such cases, the simple arithmetic mean is inappropriate, because it gives the wrong answers. What we need to find is the geometric mean, simply called the G.M.

Number of x values


$$\text{G.M.} = \sqrt[n]{\text{product of all } x \text{ values}}$$

- Suppose that over 5 years banks pay interest at annual rates of 100, 200, 250, 300, and 400 percent, which correspond to growth factors of 2, 3, 3.5, 4, and 5.

- In 5 years, an initial deposit of \$100 would grow to

$$\$100 \times 2 \times 3 \times 3.5 \times 4 \times 5 = \$42,000.$$

- The arithmetic mean growth factor is $(2 + 3 + 3.5 + 4 + 5)/5$, or 3.5. This corresponds to an average interest rate of 250 percent. Yet if the banks actually gave interest at a constant rate of 250 percent per year, then \$100 would grow to \$52,521.88 in 5 years:

$$\$100 \times 3.5 \times 3.5 \times 3.5 \times 3.5 \times 3.5 = \$52,521.88.$$

- This answer exceeds the actual \$42,000 by more than \$10,500, a sizable error.
- Use GM to determine the correct growth factor:

$$\begin{aligned} \text{G.M.} &= \sqrt[5]{\text{product of all } x \text{ values}} \\ &= \sqrt[5]{2 \times 3 \times 3.5 \times 4 \times 5} \\ &= \sqrt[5]{420} \\ &= 3.347 \leftarrow \text{Average growth factor} \end{aligned}$$

- The growth in bad-debt expense for Johnston Office Supply Company over the last few years follows. Calculate the average percentage increase in bad-debt expense over this time period. If this rate continues, estimate the percentage increase in bad debts for 1997, relative to 1995.

1989	1990	1991	1992	1993	1994	1995
0.11	0.09	0.075	0.08	0.095	0.108	0.120

Median

- The median is a measure of central tendency different from any of the means we have discussed so far.
- The median is a single value from the data set that measures the central item in the data.
- This single item is the middlemost or most central item in the set of numbers. Half of the items lie above this point, and the other half lie below it.

Calculating the Median from Ungrouped Data

- To find the median of a data set, first array the data set in ascending or descending order.
- If the data set contains an odd number of items, the middle item of the array is the median.
- If there is an even number of items, the median is the average of the two middle items. In formal language, the median is

Number of x values


$$\text{Median} = \left(\frac{n + 1}{2} \right) \text{th item in a data array}$$

TABLE 3-10 TIMES FOR TRACK-TEAM MEMBERS

ITEM IN DATA ARRAY	1	2	3	4	5	6	7
TIME IN MINUTES	4.2	4.3	4.7	4.8	5.0	5.1	9.0
				↑			
				Median			

TABLE 3-11 PATIENTS TREATED IN EMERGENCY ROOM ON 8
CONSECUTIVE DAYS

ITEM IN DATA ARRAY	1	2	3	4	5	6	7	8
NUMBER OF PATIENTS	86	52	49	43	35	31	30	11
				↑				
				Median of 39				

Calculating the Median from Grouped Data

- If we have access to data only after they have been grouped in a frequency distribution, the median may be calculated as:

$$\tilde{m} = \left(\frac{(n+1)/2 - (F+1)}{f_m} \right) w + L_m$$

- Where
 - $\tilde{m}$ = sample median
 - n = total number of items in the distribution
 - F = sum of all the class frequencies up to, but not including, the median class
 - f_m = frequency of the median class
 - w = class-interval width
 - L_m = lower limit of the median-class interval

TABLE 3-12 AVERAGE MONTHLY BALANCES FOR 600 CUSTOMERS

Class in Dollars	Frequency
0–49.99	78
50.00–99.99	123
100.00–149.99	187 Median class
150.00–199.99	82
200.00–249.99	51
250.00–299.99	47
300.00–349.99	13
350.00–399.99	9
400.00–449.99	6
450.00–499.99	2
	600

Advantages and Disadvantages of the Median

- The median has several advantages over the mean.
- Extreme values do not affect the median as strongly as they do the mean.
- The median is easy to understand and can be calculated from any kind of data—even for grouped data with open-ended classes unless the median falls in an open-ended class.
- We can find the median even when our data are qualitative descriptions.

- The median has some disadvantages as well.
- Certain statistical procedures that use the median are more complex than those that use the mean.
- Also, because the median is the value at the average position, we must array the data before we can perform any calculations.
- This is time-consuming for any data set with a large number of elements. Therefore, if we want to use a sample statistic as an estimate of a population parameter, the mean is easier to use than the median.

- Swifty Markets compares prices charged for identical items in all of its food stores. Here are the prices charged by each store for a pound of bacon last week:

\$1.08 0.98 1.09 1.24 1.33 1.14 1.55 1.08 1.22 1.05

- Calculate the median price per pound.

- For the following frequency distribution, determine the estimated value of the median.

Class	Frequency
100–149.5	12
150–199.5	14
200–249.5	27
250–299.5	58

Class	Frequency
300–349.5	72
350–399.5	63
400–449.5	36
450–499.5	18

Mode

- The mode is the value that is repeated most often in the data set.
- Sometimes chance causes a single unrepresentative item to be repeated often enough to be the most frequent value in the data set. For this reason, we rarely use the mode of ungrouped data as a measure of central tendency.

TABLE 3-13 DELIVERY TRIPS PER DAY
IN ONE 20-DAY PERIOD

Trips Arrayed in Ascending Order				
0	2	5	7	15
0	2	5	7	15
1	4	6	8	15
1	4	6	12	19

Calculating the Mode from Grouped Data

- When data are already grouped in a frequency distribution, we must assume that the mode is located in the class with the most items, that is, the class with the highest frequency.

TABLE 3-14 FREQUENCY DISTRIBUTION OF
DELIVERY TRIPS

CLASS IN NUMBER OF TRIPS	0-3	4-7	8-11	12 and more
FREQUENCY	6	8	1	5
		↑ Modal class		

- To determine a single value for the mode from this modal class

$$MO = L_{Mo} + \left(\frac{d_1}{d_1 + d_2} \right) w$$

- Where
 - L_{Mo} = lower limit of the modal class
 - d_1 = frequency of the modal class minus the frequency of the class directly above it
 - d_2 = frequency of the modal class minus the frequency of the class directly below it
 - w = width of the modal class interval

Multimodal Distributions

- What happens when we have two different values that each appear the greatest number of times of any values in the dataset.

TABLE 3-15 BILLING ERRORS PER DAY IN
20-DAY PERIOD

Errors Arrayed in Ascending Order			
0	2	6	9
0	4	6	9
1	4	7	10
1	4	8	12
1	5	8	12

- This distribution, then, has two modes and is called a bimodal distribution.

Advantages and Disadvantages of the Mode

- The mode, like the median, can be used as a central location for qualitative as well as quantitative data.
- Also like the median, the mode is not unduly affected by extreme values.
- We can use it even when one or more of the classes are open ended.
- Despite these advantages, the mode is not used as often to measure central tendency as are the mean and median.
- Sometimes, there is no modal value because the data set contains no values that occur more than once.
- Other times, every value is the mode, because every value occurs the same number of times.
- Another disadvantage is that when dataset contains more than one mode, they are difficult to interpret and compare.

- The ages of a sample of the students attending Sandhills Community College this semester are:

19 17 15 20 23 41 33 21 18 20

18 33 32 29 24 19 18 20 17 22

55 19 22 25 28 30 44 19 20 39

- Construct a frequency distribution with intervals 15–19, 20–24, 25–29, 30–34, and 35 and older.
- Estimate the modal value.

Mean, Median, And Mode Comparison Table

	Aspect	Mean	Median	Mode
1	Definition	The average of all values in a data set, found by adding them up and dividing by the number of	The middle value in a data set when it is arranged in ascending or descending order.	The value that appears most frequently in a data set.
2	When to Use	When data is evenly distributed without outliers (extremely high or low values).	When data is skewed or contains outliers that could distort the mean.	When you are working with categorical data or want to identify the most common value.
3	Best For	Finding the overall average, especially for continuous data (e.g., heights, weights, test scores).	Finding the middle ground in skewed data (e.g., house prices, incomes).	Finding the most frequent value (e.g., most popular product size, color, or category).
4	Strengths	Represents all data points; easy to calculate for most sets of numbers.	Not affected by outliers, giving a better idea of what's typical in skewed data.	Works well with categories and highlights the most popular or common item.
5	Weaknesses	Can be skewed by outliers, giving a misleading picture if extreme values are present.	Does not take into account all values in the data set; only focuses on the middle.	May not be useful if no values repeat or if there are multiple modes.
6	Example	Average test scores in a class of students.	Median house prices in a neighborhood.	Most common clothing size sold in a store.